

Face Recognition Attendance System

Biometric Verification · Geofence Validation · Real-Time Tracking

• LIVE

face-recognition-attendance-system-xztu.onrender.com



Team: AYUSH GANGWAR · PRATHAM KUMAR · PALLAVI KUMARI · ABHISHEK YADAV · NEHA GANGWAR · ADITYA RAJ · SANJEEV MAURYA · JATIN SHARMA

[GitHub: github.com/rajpratham1/Face-Recognition-Attendance-System](https://github.com/rajpratham1/Face-Recognition-Attendance-System)

Problem Statement



Proxy Attendance

Students mark attendance on behalf of absent peers — impossible to detect manually



Manual Errors

Paper & sign-in sheets lead to data loss, miscounts, and teacher workload



No Audit Trail

Zero accountability — no way to track who marked, when, or from where

Our Objective

- ✓ Eliminate proxy attendance using face biometrics
- ✓ Enforce campus geofence for location validity
- ✓ Provide full audit logs for every attendance attempt
- ✓ Automate reporting with CSV exports
- ✓ Support multi-role: Student, Teacher, Admin

Key Features



Face Verification

face-api.js runs in-browser. Descriptor matching ensures only the registered student can mark attendance.



Geofence Validation

GPS coordinates verified server-side via Geopy. Students outside campus radius are auto-rejected.



Security Layers

CSRF tokens, script hashing, rate limiting, device hash & IP logging on every attempt.



Audit & Reporting

Full logs per attempt with reason, timestamp, IP. Teachers export CSV reports per course.



Role-Based Access

Three roles — Student, Teacher, Admin — each with tailored dashboards and permissions.



Email Notifications

Automated email alerts dispatched on attendance confirmation via integrated email service.

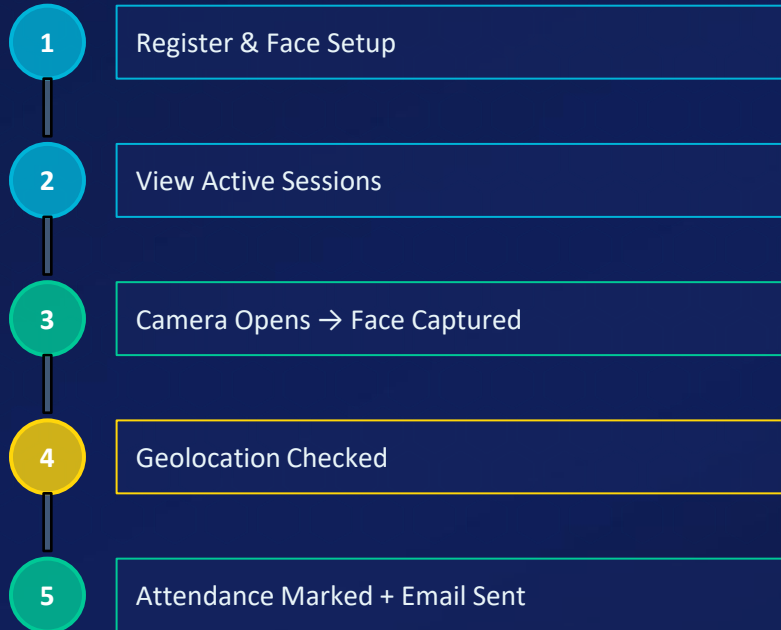
Tech Stack & Architecture

Frontend	Backend	Database	Infrastructure
▸ HTML5 / CSS3 / JS ▸ face-api.js (AI model) ▸ Jinja2 Templates ▸ Bootstrap UI	▸ Python 3.9+ ▸ Flask Framework ▸ Flask-Login / WTF ▸ Flask-Limiter	▸ SQLAlchemy ORM ▸ SQLite (local) ▸ Firebase Realtime DB ▸ Encrypted Backup DB	▸ Render.com (Live) ▸ Gunicorn WSGI ▸ Geopy Geofencing ▸ Cryptography lib

Browser (face-api.js) → Flask API → SQLAlchemy → SQLite / Firebase → Render.com

System Workflow

Student Flow



Teacher Flow



Meet the Team



AYUSH GANGWAR

LEADER



PRATHAM KUMAR

CO-ORDINATOR



PALLAVI KUMARI

CO-LEADER



ABHISHEK YADAV

DEVELOPER



NEHA GANGWAR

DEVELOPER



ADITYA RAJ

DEVELOPER



SANJEEV MAURYA

DEVELOPER



JATIN SHARMA

DEVELOPER

Live Demo & Resources



Live Deployment

face-recognition-attendance-system-xztu.onrender.com

Hosted on Render.com · Always-on via Gunicorn WSGI



GitHub Repository

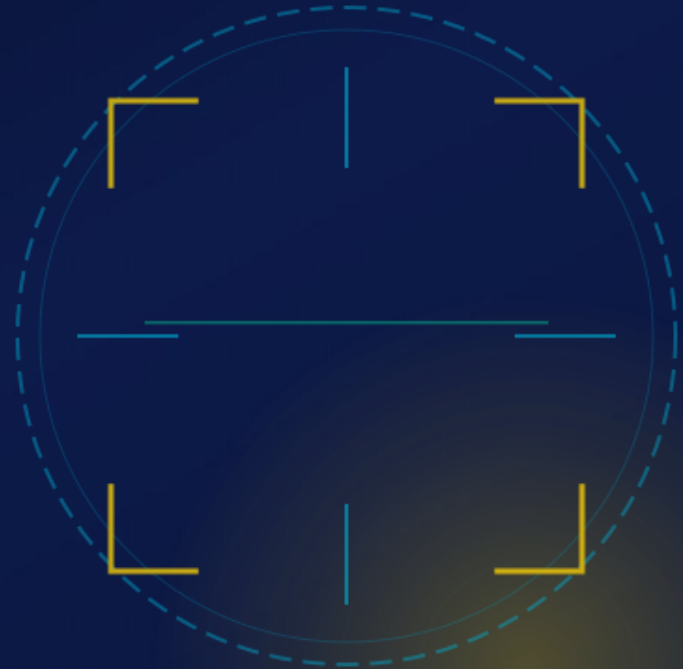
github.com/rajpratham1/Face-Recognition-Attendance-System

Full source · Documentation · Changelog · CONTRIBUTING.md



Final Project Report (PDF)

drive.google.com/file/d/1U8Gqz89wH33zmmrZoRt-m_5qpKSRBHFT/view



Thank You!